

SW06 - From Time Series to Images

In common:

- Both signals (information, patterns, noise, continuous/discrete)
- Can be processed with similar techniques
 - Filtering
 - Fourier-Transform
 - Feature extraction
- Local structure matters
 - in time-series: neighboring time points are often correlated
 - in images: neighboring pixels are often similar

Differences:

Feature	Time Series	Image
Signal dimension	1D signal varying over time: $x(t)$	2D signal varying over space: $I(x,y)$
Axis meaning	Time (temporal axis)	Spatial position (x, y)
Sampling	Typically uniform sampling in time	Discrete grid sampling (pixels) over space
Directionality	Inherently directional (past $\rightarrow$ future, causality)	No inherent directionality; spatially symmetric
Types of patterns	Periodicity, trends, temporal events	Edges, textures, shapes, regions
Visualization	Line plot (amplitude vs. time)	Image display (intensity or color map)

▷ Domain shift: From time domain $\rightarrow$ spatial domain

we move from analyzing how signals change over time to how they change across space

Topics

- Discretization in 2D
 - Sampling
 - Quantization
- Color spaces
 - Binary / RGB / RGBA / HSL / CMYK ...
 - Conversions
- Basic image operations
 - Cropping
 - Spatial image transformations
- Spatial Fourier analysis
 - (Spectrograms in 2D)
 - FFT for images
- Convolution / filtering / kernels
 - Noise removal / smoothing
 - Gradients and image derivatives
- Features
 - Edge detection
 - Corners / interest points
 - Invariance properties
- Masks / morphological operations
 - Thresholding
 - Contours
 - Erosion / dilation / closure / opening ...

SW06 - Digital Image Processing 1

Objective: Build machines that can see 🤖

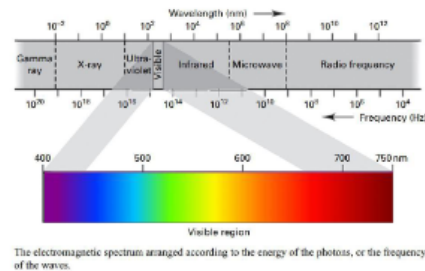
- Acquire and represent image data
- Transform and filter images
- Enhance image data, or restore missing data
- Optimize information content (compression)
- Extract information from image data (sensing)
- Understand image content (semantics)
- Decide based on image data
- Synthesize images

▶ The more complex the task, the more machine learning or AI is involved

Context: Imaging systems by type of radiation (or field) used

aw

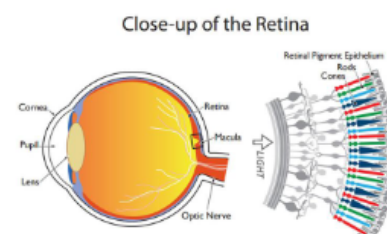
- Electromagnetic waves carry the information captured in images.
- General rules:
 - The shorter the wavelength of a wave, the better it can reveal small details
 - The shorter the wavelength, the more energy carries a wave
- Types of radiation or fields used in imaging:
 - Electromagnetic waves: Radio, microwaves, infrared, visible light, ultraviolet, x-rays (soft or hard), γ-rays
 - Particle radiation: Neutrons, protons, electrons, heavy ions
 - Mechanical (acoustic) waves: Ultrasound (medical imaging), sonar (underwater imaging), seismic waves (geophysical imaging)



Human visual perception

av

- Up to about 60% of the brain activity are involved in processing visual stimuli
- More surprising facts about the human visual system
 - The image on the retina is upside-down
 - Retina is front-back inverted
 - Peripheral vision is low-res and nearly monochrome
 - The resolution of the eye is only 1 megapixel
 - The eye involves about 100 million photoreceptors
 - The eye has a special night-vision mode



Human visual perception

aw

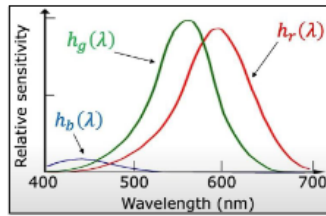
3

- Rods and cones are specialized photoreceptor cells located in the retina of the eye.
 - Rods: very sensitive to light in general, particularly in low-light conditions, provide peripheral vision
 - Cones: Less sensitive to light, but provide the ability to perceive colors, enable high visual acuity (Sehschärfe)

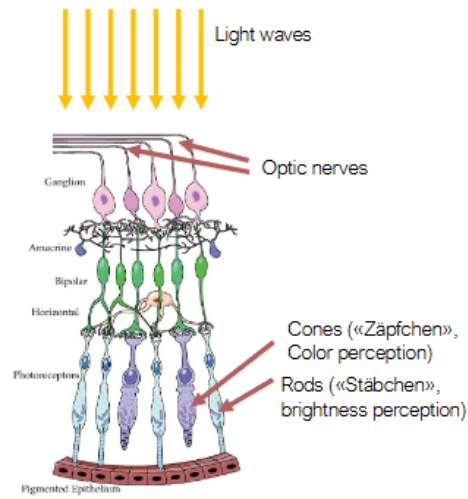
Three types of cones with different spectral sensitivity:

- S-cones / blue
- M-cones / green
- L-cones / red

S: Short-wavelength
M: Medium-wavelength
L: Long-wavelength



«Quantum efficiency» of red, green and blue cones.



av

Luminance vs. Chrominance

The human visual system processes **brightness** and **color** differently.

▪ Luminance: Brightness perception

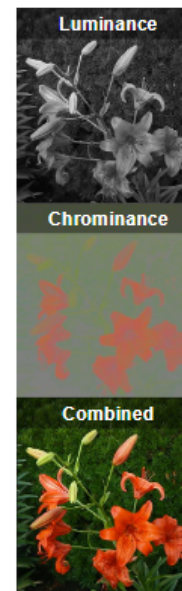
- Perceived light intensity of a scene
- Mainly processed by rods
- High sensitivity to contrast, edges, fine spatial details

⇒ Dominates our perception of **structure** and **sharpness**

▪ Chrominance: Color perception

- Represents color information (hue and saturation)
- Processed by cones
- Lower spatial resolution, less sensitive to fine detail

⇒ Adds **color appearance**, but contributes less to sharpness

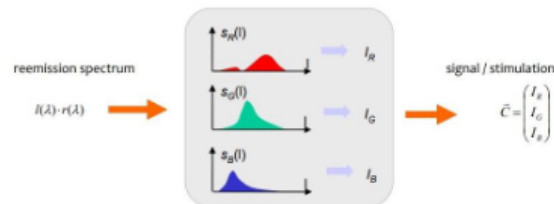


Color perception

aw

▪ Light propagation:

- Light source spectrum: $I(\lambda)$
- Reflectance / reemission spectrum: $r(\lambda)$
- Sensitivity of cone receptors: $s_x(\lambda)$



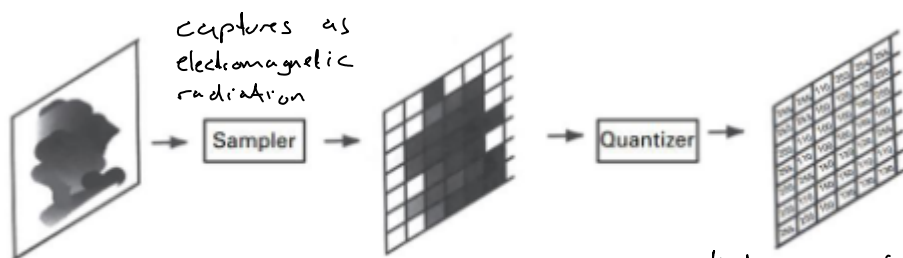
▪ Result:

- Signal / stimulation consists of three channels
- Different stimulation patterns lead to the perception of different colors:



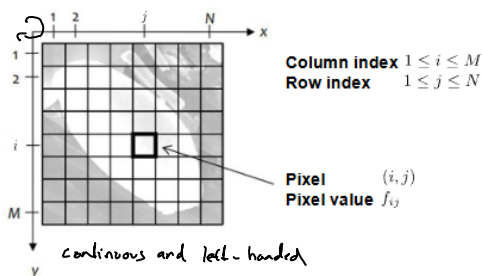
Digital Image Processing II

4



The relationship between an analog image and a digitized image.

digital arrays of discrete values



Matrix representation

$$B = (f_{ij}) = \begin{pmatrix} f_{11} & f_{12} & \cdots & f_{1N} \\ f_{21} & f_{22} & \cdots & f_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ f_{M1} & f_{M2} & \cdots & f_{MN} \end{pmatrix}$$

$\Rightarrow$ right-handed

$\Rightarrow$ right-handed

how many grayscale levels?

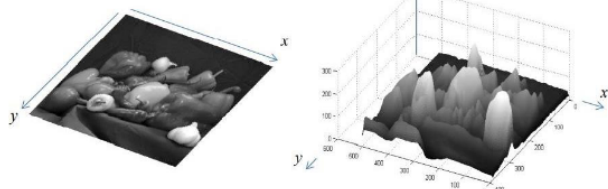
- Image depth (bit depth)
 - nr of bits = intensity or color of each pixel
 - how display shades
- Integer vs. floating-point representation
 - stored as integers (8-bit, 16-bit ...)
 - image processing as floating-point (16, 32, 64 bit)
 - normalization $[0, 1]$
- Other bit depths and representations
 - 16 bit images in scientific and medical images for higher dynamic range
 - lower bit depths as 8-bit to reduce memory
 - floating-point used during processing

Image as a function

- A painted grayscale image can be considered as a continuous function over a region in $\Omega \in \mathbb{R}^2$, then at each point (x, y) point we get a gray value g which we can describe by a number $g = f(x, y) \in \mathbb{R}$:

$$f : \Omega \in \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \rightarrow f(x, y)$$

$f(x, y) \Rightarrow$ grey-value



8 bits (1 byte) gray images corresponds to 256 possible gray levels

0 ... 0 $\Rightarrow$ black
1 ... 1 $\Rightarrow$ white

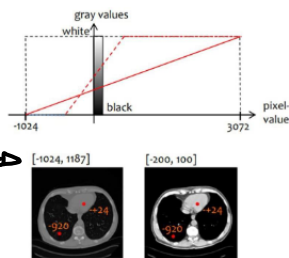
Pixel values / Grayscale values

- In many applications, pixel values represent physical quantities, not just brightness (unlike in standard grayscale photography). For example:

- Radiation absorption (e.g., X-ray / CT)
- Magnetization (e.g., MRI)
- Other modality-specific signals

Implications:

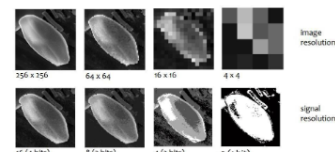
- Pixel values may lie in arbitrary value ranges (and not 0-255)
- For storage and visualization, we still prefer a standard value range $[0, 255]$
- Therefore, pixel values must be mapped (normalized) to this range



In CT data, pixel values typically range from -1024 to 3072. If only a selected portion of this range is mapped to the display grayscale range, this process is called windowing.

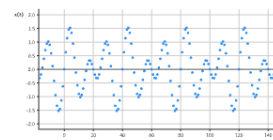
Digitalization / discretization

- Digitalization involves two steps:
 - Spatial quantization (sampling)
 - Intensity quantization.
- Quantization means that values are restricted to a finite set of permissible levels (e.g., integers instead of continuous values)



Why 8-bit images are common

- The human visual system can distinguish only a limited number of gray levels (≈ 80 shades, depending on conditions)
- 8-bit (256 levels) is therefore sufficient for most visual applications



Task: Try to connect this to time-series signals.

Intensity quantization

- In grayscale images, each sampled intensity value must be assigned to the nearest available gray level
- This introduces an approximation error, known as **quantization error**
- The error decreases as the number of gray levels increases
- However, more gray levels require higher bit depth and therefore more storage

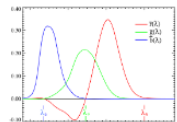
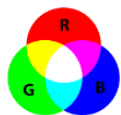


64 shades of gray

8 shades of gray

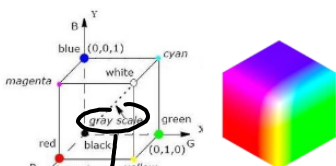
The RGB color space

aw Trichromatic theory



These curves do not match the spectral sensitivity curves of humans for various reasons (e.g., color light generation in 1931)

Human color perception can be approximated by combining 3 primary colors red, green and blue



• diagonal axis, represent the gray scale

Example: RGB color space

- In an RGB color image, each pixel contains three values (R, G, B); therefore, a color image can be represented as three grayscale images.



$$p = \begin{pmatrix} 255 \\ 255 \\ 0 \end{pmatrix}$$



Task: Can you guess which of the below images represent R, G or B?

The HSI color space (Hue, Saturation, Intensity)

Limitations of RGB:

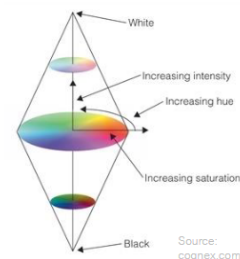
- RGB is well suited for image acquisition and display (cameras, monitors)
- However, interpreting colors in RGB is not intuitive.

Motivation:

- Represent color in a way that aligns with human perception
- Separates color information (H, S) from brightness (I)

Components of HSI:

- H: Hue (type of color, e.g., red, yellow, magenta)
- S: Saturation (purity or vividness of the color)
- I: Intensity (brightness of the color)



Source: cognex.com

The HSI color space (Hue, Saturation, Intensity)

- In the HSI space, the color can be manipulated without affecting the brightness or saturation of color.



Other color spaces

CMYK (Cyan, Magenta, Yellow, Key / black):

- Represents colors by subtracting cyan, magenta, and yellow from white light.
- Hence, it is a subtractive color model (contrary to RGB, or HSI)
- Includes a black channel (K, key) to improve the reproduction of dark colors
- Standard color model used in color printing and graphic design.



L*a*b* (CIELAB):

- A device-independent color space designed to approximate human vision more accurately than RGB or CMYK.
- Consists of three components: L* (lightness), a* (green to red), and b* (blue to yellow).
- Widely used in color science and for color difference calculations



XYZ (CIE 1931 XYZ):

- A standard based on human color perception
- Represents colors using tristimulus values X, Y, and Z, which correspond to the stimulation of the three types of cones in the human eye.



Reduction of color spaces: Dithering

- Like the gray scale, also the color scale can be reduced
- Simple nearest-neighbor color-mapping yields color bands
- Dither: Method to avoid such pattern by adding noise



Method 1: Nearest neighbor mapping results in color bands and a comic-like appearance.



Method 2/3: Introduce random noise to avoid bands (second method: use diffusion approach to for better results)



Input: $2^8 \cdot 3 = 2^4 = 16.7$ Mio color
Goal: 10 target colors



Reduction of color spaces: Color to grayscale conversion

Goal: convert the three RGB value (per pixel) to one grayscale intensity value

Input image



value



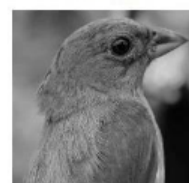
$$V = \max(R, G, B)$$

luminance



$$L = \frac{\max(R, G, B) + \min(R, G, B)}{2}$$

intensity



$$I = \frac{R + G + B}{3}$$

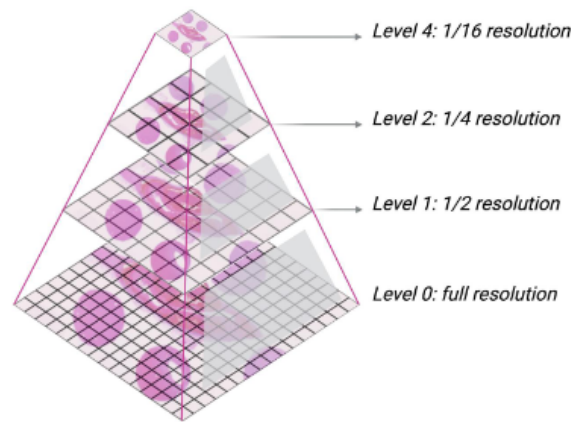
weighted intensity



$$I = 0.3 \cdot R + 0.6 \cdot G + 0.1 \cdot B$$

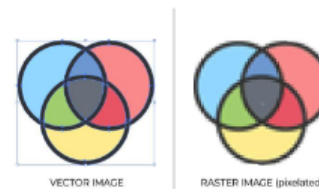
Alternative image representations: Pyramidal images

- Alternative data structure for images
- Hierarchical structure of images created by successive subsampling operations (and prior filtering)
- Idea: Maintain different resolutions of the same image
- Advantages:
 - Memory optimization: Only load parts into memory that are currently required
 - Enables multi-resolution analysis
- Disadvantage:
 - Requires more memory

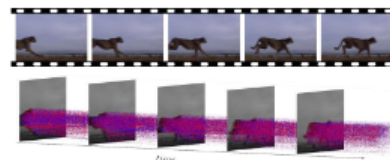


Alternative image representations

- Vector graphics (instead of rasterized data)
 - Instead of representing an image as a grid of pixels, describe it using geometric primitives such as points, lines, curves, or polygons.
 - More on this later...
- Sparse / event-based representations
 - Not all images require a dense pixel grid.
 - Sparse images store only relevant or non-zero pixels.
 - Event-based cameras capture changes over time → Data is stored as a list of events, not a full matrix.
 - [Explanatory video](#), by LUCID Vision Labs



Source: impact-nw.com



Event-based vs. dense imaging: only temporal changes are recorded, enabling higher effective sampling rates. Source: [Wikimedia](#)

aw

The digital image processing system

A complete digital image processing system consists of hardware (equipment) and software (algorithms and programs) that together perform the following tasks:

1. **Image acquisition:** Capture an image and, if necessary, convert it to digital form using an analog-to-digital converter (ADC)
2. **Storage:** Store the image temporarily (in memory) or permanently (on disks or SSDs)
3. **Processing:** Manipulate and analyze the image (e.g., enhancement, filtering, segmentation)
4. **Display:** Visualize the image, typically via a display system (historically involving DACs)

